Can Direct Micropayment Incentives Shape the Quality of Online Discourse?*

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Abstract

This paper studies how small posting fees and peer-to-peer monetary rewards shape the quality and quantity of posts in online discourse. The setting is an online discussion platform that uses Bitcoin micropayments. On the platform, users must pay a small fee to post content and can earn peer-to-peer tips from other users who value their contributions. Using variation in posting fees across the platform's sub-forums and over time, we estimate how post quantity and quality change after a posting fee change. We find that the number of posts responds negatively to a posting fee increase, with an elasticity of -0.265, but that the quality of posts, as measured by tips earned, responds positively to a posting fee increase, with an elasticity of 0.219. These results are consistent with a screening model in which posting costs filter out low-quality contributions. We also find evidence that users adjust subsequent posting behavior in response to observed tip-based feedback, and that users who do not recover their posting costs through tips are more likely to exit the platform. These findings provide empirical evidence on how small financial incentives can shape the quantity and quality of online discussion.

Keywords: micropayments; user-generated content; signaling; market design; digital platforms
JEL Classification: D82, D83, L86

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1 Introduction

Can direct financial incentives shape the quality of online discourse? A large literature has studied how indirect mechanisms—such as ad revenue sharing, algorithmic visibility, or reputation systems—influence the quality and quantity of content that users produce (Kraut et al., 2012; Luca, 2015; Aridor et al., 2024). However, direct financial mechanisms such as requiring users to pay a fee to post or allowing peer-to-peer payments between users are rare in the field and thus remain largely understudied. This is especially true in the market for online discussion platforms, since each individual post in a discussion platform like Reddit or X is typically of sufficiently low value that it is not worth the transaction costs to introduce payments into these systems.

Yet, there is good reason to believe that direct payment mechanisms could meaningfully influence user behavior and content quality on discussion platforms. Signaling theory suggests that posting costs can serve as a screening device, deterring low-quality contributions while allowing users with high-quality content to signal their type by paying the cost (Joseph and Thevaranjan, 2008). Empirically, Tchernichovski et al. (2019) showed that introducing a small fee to an online rating system filtered out low information signals and increased the correlation between review ratings and quality. Other empirical studies have found that financial incentives in the form of revenue sharing can influence the quantity and quality of user-generated content (Sun and Zhu, 2013; El-Komboz et al., 2023; Kerkhof, 2024). However, none of these studies have investigated the behavioral effects of posting fees and direct peer-to-peer payments in a naturally occurring discussion platform.

One reason direct payment mechanisms have not been more widely adopted is frictions in the payments infrastructure. The value of any individual contribution is typically too small to justify credit card transaction fees, and international platforms face additional frictions related to currency exchange and cross-border banking regulations. However, the emergence of low-cost digital payment rails, such as Bitcoin's Lightning Network, has made it easier for platforms to integrate micropayments into their core design, opening the door to incentive structures that were previously impractical.[1]

This paper provides empirical evidence on how micropayment incentives shape user behavior, using data from a Bitcoin-based discussion platform called Stacker News. Stacker News is an internet discussion forum—similar in format to Hacker News or Reddit—that integrates Bitcoin micropayments into its core design through the Lightning Network. Users must pay a fee denominated in Bitcoin, usually on the order of 1-100 satoshis (about 0.1 to 10 cents at the time the research was conducted), to post content in any of the platform's topic-specific sub-forums, called territories.[2] Users can also send each other peer-to-peer Bitcoin micropayments, called "zaps", to reward each other for content that they value. The combination of posting fees and peer-to-peer tipping creates a setting in which both the cost and reward sides of content production are mediated by real financial transactions, making Stacker News a natural field setting for studying how direct payment incentives influence user behavior on discussion platforms.

We exploit variation in posting fees across territories and over time to estimate how the quantity and quality of posts change in response to a territory fee change. Using a

[1]The Lightning Network is a decentralized payments network that is built on top of Bitcoin. It enables faster and lower cost Bitcoin transactions than the base layer would allow. It works by letting users maintain peer-to-peer payment channels that allow them to "float" Bitcoin between each other indefinitely, like keeping a bar tab, until final settlement occurs on the base layer. Users can pay others with whom they have no direct channel by routing payments through other channels, usually at a small percentage fee.

[2]A satoshi (or sat) is 1/100,000,000th of a Bitcoin. At the time the research was conducted, 1,000 sats was worth about USD $1.

difference-in-differences framework with territory and time fixed effects, we find that doubling the posting cost in a territory reduces the number of posts by 16.8%, but increases the quality of posts—measured by tips received from other users—by 16.4%. Because territory operators can change posting fees endogenously, a central empirical concern is whether operators tend to change fees during periods when quantity is already trending down or when quality is already trending up. We conduct event study and pre-trends analysis to verify that this is not the case; indeed, the behavioral response follows the fee change rather than precedes it. The results are consistent with a screening model in which users with low-quality contributions are unwilling to pay the higher cost to post, while users with high-quality contributions are willing to pay the higher cost.

Beyond the cost-based screening mechanism, we investigate whether users learn from and respond to observed tip feedback. To do this, we grouped the posts into different types and measured the expected tip reward for each type under different assumptions about each user's information environment. We find that users are more likely to make a post of a given type when the expected tip reward for that type is high relative to the expected tip rewards for the other types. In other words, users respond to observed tip feedback by shifting the posts they make towards better-rewarded types. The effect is moderated by the user's experience level. More experienced users respond less sharply to observed tip feedback, suggesting that learning takes place early when their beliefs are more diffuse, consistent with prior work on social learning (Conley and Udry, 2010). Moreover, we find that users respond more sharply to tip feedback on their own posts than to feedback on other users' posts, suggesting that feedback on their own posts is more salient than feedback on other users' posts. In the aggregate, we find that the composition of post types has shifted towards the better rewarded types over time, thus showing the effect of tip-driven learning on aggregate post quality.

Lastly, we investigate whether financial incentives shape the composition of active users over time. We find that users who are unable to earn enough tips to cover their posting costs are more likely to exit the platform, while users who earn more tips than they pay in posting fees are more likely to stay. This results in a temporal selection mechanism which could further improve content quality over time.

Our findings are both surprising and unsurprising. They are unsurprising in the sense that the behavioral responses to the financial incentives observed are exactly what economic theory would predict. Demand curves slope downwards, signaling theory works, and people learn from feedback. The results are surprising in the sense that the behavioral responses are detectable and robust for even very small financial incentives, on the order of 5 to 10 cents, in a natural field setting.

Our findings have policy implications for designers and regulators of user-generated content platforms. The results suggest that content quantity and quality are mediated by financial incentives, even very small ones. Direct costs to participate can be used to screen out low quality content, while peer-to-peer tipping incentives the production of higher quality content. While our setting is a single platform with a highly selected user community, the underlying mechanisms we identify—cost-based screening, tip-driven learning, and profitability-based selection—are general features of micropayment systems and could plausibly operate on other platforms that adopt similar designs.

2 Incentive structures on Stacker News

From its outset, Stacker News was founded with the idea of using Bitcoin micropayments to incentivize quality content and good behavior. Its motto is “Stacker News is trying to fix online communities with economics." This makes it a natural field setting for studying how micropayment incentives shape user behavior. There are a number of incentive structures built into the design of Stacker News.[3]

Direct posting cost. First, users must pay to post. The posting cost is denominated in sats and varies by territory. A territory is a sub-forum on Stacker News, usually topic-specific.[4] Territories are created and operated by individual users on Stacker News. To create a territory, a user chooses a name for the territory, writes a description, and pays a monthly fee to Stacker News to maintain the territory. As of October 2025, the monthly fee to operate a territory is 50,000 sats (about $50 a month at the time the research was conducted).

Once a new territory is founded, users can post in that territory by paying a posting fee which is set by the territory operator. The territory operator earns 70% of the posting fee as revenue, and the other 30% is "taxed" and later redistributed as part of a daily rewards pool.[5] Territory operators are therefore incentivized to choose a posting fee that optimizes between post quantity, post quality, and revenue earned. If the posting fee is too low, the territory may not earn enough revenue to cover costs, and it may also attract low quality posts. On the other hand, if the fee is too high, users may be deterred from posting in the territory. Users are incentivized to post in the territory that best fits the topic of their post so as to maximize engagement and sats earned net of posting fees.

[3] The following description of how Stacker News works is accurate up to October 5, 2025.

[4] As of October 2025, the most popular territories are `bitcoin`, which focuses on discussions about Bitcoin; `econ`, which focuses on news and discussion about economics; `Stacker_Sports`, which focuses on sports discussion; `AI`, focused on news and discussion about artificial intelligence; and `Politics_And_Law`, focused primarily on politics and current events. The popularity of these territories shows that Stacker News is not a Bitcoin-specific message board but covers a wide range of topics.

[5] The purpose of the tax is to discourage spam. Without the tax, territory operators would be able to post costlessly and without limit in their own territories. Revenues collected from this tax are distributed each day to active users according to an algorithm that is opaque to most users.

Peer-to-peer tipping. Second, users can tip each other directly for content that they like. On Stacker News, these tips are called "zaps". To zap an item, the user clicks a lightning bolt icon which is featured prominently on every post and comment. When the user clicks the lightning bolt, sats are sent from the user to the creator of the item. 70% of the zap amount is sent directly to the item creator's wallet, while 21% are sent to the territory operator and 9% are sent to the daily rewards pool.[6] Users can set a default zap amount, but they can also customize the amount they want to zap any particular post. Henceforth, we shall use the words "zap" and "tip" interchangeably.

Like Reddit upvotes, zaps are used by the platform to surface high quality content—the default view on Stacker News ranks posts based on the amount of zaps earned by the post, modified by a time decay factor to keep the view fresh. Unlike Reddit upvotes, zaps represent real monetary value transferred from the zapper to the creator of the post. The zapping system incentivizes high quality content in two ways: not only does the system surface higher quality posts to the top of the home page, but it also lets creators of high quality content get paid directly by other users for their work.

While it is clear how zapping incentivizes content creators, it is not as clear what incentives users have to zap. Economists who have studied tipping in non-digital contexts have argued that tipping is not well explained by considerations of future value (Azar, 2020) and have instead focused on altruistic "warm glow" effects (Andreoni, 1990; Crumpler and Grossman, 2008). However, these are contexts where interactions are generally not repeated and where the payment includes a deterministic price along with an optional tip. On Stacker News, users have repeated interactions with each

[6] The 30% tax on zaps is a form of sybil resistance: it discourages users from creating sock puppet accounts to zap their own content.

other and there is no fixed financial reward for content creators other than tips. The incentive model may therefore be closer to that of a "pay-what-you-want" system, in which tipping is rationally used to ensure that producers do not switch to a less advantageous pricing model for consumers (Mak et al., 2015). In the case of Stacker News, the consequence of not tipping could be a reduction in content production that the user enjoys. Whether users zap due to an altruistic desire to reward content that they enjoy, or by a rational incentive to support the continued production of such content, or by a desire to influence the ranking of posts, we do observe a healthy amount of zapping on Stacker News.

Hypotheses. Based on the above discussion, we hypothesize the following about the incentive structures on Stacker News. First, we hypothesize that direct costs to post will screen out low quality content. Thus, when a territory increases its posting fee, the quantity of posts should decrease but the quality of posts should increase. Second, we hypothesize that users learn from and respond to tip feedback. Thus, users will shift their posting behavior towards the types of posts that were observed to earn the highest rewards. Third, we hypothesize that users who are consistently unable to cover their posting fees with tips are more likely to leave the platform. We test these hypotheses one-by-one in the following sections of the paper.

3 Data

To test these hypotheses, we downloaded a snapshot of the Stacker News database on October 5, 2025. The data covers the time period from June 11, 2021 to October 5, 2025 and contains 207,219 posts and 914,266 comments (replies), for a total of 1,121,485 items posted by 11,228 users across 117 territories. Over this time period, users zapped each other 146 million sats and paid 13 million sats in posting fees (about $146,000 and $13,000 respectively, at the time the research was conducted).

In addition to the Stacker News data, we downloaded daily Bitcoin price data from CoinMarketCap.com. At the time the research was conducted, the price of one Bitcoin was approximately US$100,000, which would make 1,000 sats worth approximately $1. This is a useful rule of thumb for converting between sats and dollars when trying to evaluate the economic significance of Stacker News micropayments.[7]

Figure 1 shows the growth in Stacker News, as measured by weekly number of items posted, alongside the weekly Bitcoin price. In aggregate, both the Bitcoin price and the activity on Stacker News grew over this time period.

4 Cost-based screening

In this section, we consider our first hypothesis: whether higher posting costs lead to fewer posts in a territory, but higher quality posts.

4.1 Evidence on post quality

To test the relationship between posting fee and post quality, we run regressions of the following form:

$$\ln \text{Quality}_{ijt} = \alpha \ln \text{Cost}_{ijt} + \delta_j + \gamma_t + \epsilon_{ijt} \tag{1}$$

where i indexes a post, j indexes a territory, and t indexes a year-week.[8] Quality_{ijt} is the measure of post quality, Cost_{ijt} is the cost (in sats) of the post, δ_j is a set of

[7] In the subsequent text, whenever we convert sats to dollars, we will be using a conversion rate of 1,000 sats = $1.

[8] For example, 2024w1 is a different t than 2025w1.

dummy variables for each territory, and γ_t is a set of dummy variables for each week. The coefficient α identifies the elasticity between posting cost and post quality.[9] We hypothesize that $\alpha > 0$.

To measure post quality, we measure the number of sats the post earned on zaps in the first 48 hours of posting. Zaps represent a revealed preference measure of quality, since users must pay real money to zap. The purpose of the dummy variables is to absorb any unobserved heterogeneity in territories (e.g. if some territories are more popular and thus attract more zaps), and to absorb any global time trends in the popularity of Stacker News or the generosity of its users.

The effect of posting cost is identified from differential changes in post quality across territories when the territory posting costs change differentially. For example, if the posting cost in the `econ` territory increases while the posting cost in `bitcoin` stays the same, and subsequently the average quality of posts increases in `econ` relative to that in `bitcoin`, this would suggest that the change in posting cost likely caused the change in post quality. The coefficient α is identified off the many such changes present in the data.

To give the reader a sense of the variation in posting costs, we plot the posting cost histories for four representative territories in Figure 2. The figure shows that there is indeed substantial variation in posting fees both across and within territories.

Before running the regression, we apply the following sample selection criteria. We do not include posts in territories that are run by Stacker News itself for special purposes.[10] We also do not include a number of special post types that do not follow

[9] In practice, we add 1 to post quality and posting cost to avoid taking logs of zero. Since the means of both variables are well above one, the $\ln(1+x)$ transformation closely approximates $\ln(x)$, and the coefficients retain their interpretation as elasticities.

[10] This includes territories like the `jobs` territory, which companies (usually Bitcoin-focused companies) can make job postings, or the `AMA` territory, where influential people are invited to make "Ask Me Anything" posts.

normal posting rules.[11] Lastly, we do not include any posts made in territories by the territory operator. Since 70% of posting costs go to the territory operator, the incentives for territory operators to post in their own territories differ from that of non-operators. In the data we use for the regression, the average posting cost is 51 sats, the average amount of sats earned in the first 48 hours is 364 sats, and the average number of comments in the first 48 hours is 3.3. There is substantial variation in all three: the standard deviation of the posting cost is 241 sats, of sats earned in the first 48 hours is 6,662, and of comments in the first 48 hours is 9.6.

Table 1 shows results for regressions of specification (1) where post quality is measured as zaps in the first 48 hours. Each column of Table 1 includes a different set of fixed effects in the regression. Column 1 includes no fixed effects, and is therefore a straight OLS regression of log zaps in the first 48 hours on log posting cost. Column 2 includes territory and week fixed effects to control for any heterogeneous effects between territories. Column 3 includes fixed effects for the identity of the territory operator to control for any differential zapping behavior between territory operators. For example, some territory operators zap every post that comes into their territory while others do not. Territory-operator fixed effects control for this behavior.[12] Finally, column 4 includes user fixed effects to test whether the gains in post quality are driven by within-user effects or compositional effects.

Table 1 shows that there is a statistically significant and robust positive relationship between territory posting fees and post quality as measured by zaps in the first 48 hours. In terms of the magnitude of the effect, Table 1 column 3 (our preferred specification) suggests that the elasticity between territory posting cost and expected

[11]These include profile posts, which users can post and edit for free, "saloon" posts, which are daily discussion threads posted by Stacker News, and freebies—a limited number of reduced-visibility posts that users are allowed to make for free.

[12]Note that territory fixed effects do not fully absorb territory operator fixed effects, since a single user can own multiple territories, and territories can also change operators over time.

post quality is 0.219. This roughly translates to: if a territory doubles its posting cost, then the quality of posts is expected to increase by 16.4 percent. The relationship is not explained by territory specific effects, such as a spurious correlation between posting fees and the popularity of a territory's topic, nor is it explained by global time trends in posting fees and zap behavior on Stacker News. Neither is it explained by effects related to territory operators—for example, if territory operators who set high fees also tend to zap posts in their territory more generously. The best explanation for the correlation between posting cost and post quality is that users will only choose to post in a higher cost territory if they think that the post is of high enough quality to warrant the cost. This is consistent with users' self-reported perceptions of their own behavior on the platform. One Stacker News user is quoted saying:

> "If I put a lot of effort into a post, I know that the posting fee will be offset by the zaps. And even if it doesn't, I'm just happy to show the work I've done and I'm ready to pay for it. On the other hand, when I share a simple link with a few quick comments and quotes, it takes me a few minutes, and I look for the cheaper territory to post it. If I don't find any cheap territory, I'd rather not post, as I don't think paying to post links is worth the same as paying to post for [my] actual proof-of-work."[13]

Finally, we comment on column 4 of Table 1, where we include user fixed effects. The inclusion of user fixed effects reduces the estimated elasticity between posting cost and post quality from 0.219 to 0.091 (still statistically significant at the 5% level). This suggests that the increase in post quality in response to an increase in posting cost is driven by both intensive margin (within-user) and extensive margin

[13] In this statement, the user is referring to link posts, which are posts that include a link to an external URL, often without any text in the post body. Link posts usually receive less zaps and are considered to be of lower quality because the content is not produced by the user themselves, and are thus less rewarded by zaps.

(compositional) effects. On the intensive margin, the same users now make higher quality posts, conditional on posting. On the extensive margin, the composition of posters in the territory shifts towards higher quality, most likely due to lower quality users dropping out. Column 3 remains our preferred specification over column 4, because we view the effect on the composition of users as a dimension we want to capture when studying the relationship between posting cost and post quality.

4.1.1 Event study analysis

Our empirical setting is similar to a difference-in-differences (DID) regression, because we are identifying the effect of territory fee changes from the change in post quality following the fee change, while allowing for baseline differences between territories and common time trends in post quality. In classical DID, an important assumption is that the average post quality in a territory does not systematically trend upwards prior to a post fee change. If it did so, that would call into question whether the timing of the fee change is endogenous to some prior effect that was already increasing quality. Our setting is a little more complicated than a classical DID because we have multiple fee changes in a territory, bidirectional fee changes, and continuous variation in the size of the fee change. Nevertheless, it would be useful for us to test whether there are any systematic post quality changes prior to a fee change.

To test this, we run the following regression:

$$\begin{aligned} \ln \text{Quality}_{ijt} = & \sum_{s=-12}^{12} \alpha_s^{+} 1\left[\Delta \text{Cost}_{j,t-s} > 0\right] \\ & + \sum_{s=-12}^{12} \alpha_s^{-} 1\left[\Delta \text{Cost}_{j,t-s} < 0\right] + \delta_j + \gamma_t + \epsilon_{ijt} \end{aligned} \tag{2}$$

Here ΔCost_{jt} is a continuous variable for how much the posting cost in territory j

changed in week t, and $1[\Delta \text{Cost}_{jt} \gtrless 0]$ is an indicator for whether the posting cost in territory j increased/decreased in week t. Thus, $\alpha_s^{+/-}$ measures the contribution to post quality of being s weeks away from a posting cost increase/decrease when $s < 0$, and the contribution to post quality of being s weeks after a posting cost increase/decrease when $s > 0$.

We estimate equation (2) and plot the estimates for $\alpha_s^{+/-}$, along with their 95% confidence intervals, in Figure 3. The left panels show the results for a specification in which user fixed effects were not included, and the right panels show the results for a specification in which user fixed effects were included.

Figure 3 shows that there was no systematic upward trend in post quality prior to post fee increases. After a post fee increase, we see that post quality begins to trend upwards. By comparing the top left and top right panels, we also see that including user fixed effects moderates the effect, but does not eliminate it—suggesting the presence of both compositional and intensive margin effects. Although noisier, we find a similar result surrounding decreases in post fees. There is no systematic downward trend in post quality prior to post fee decreases, and following a post fee decrease post quality begins to trend slightly downwards.

4.2 Evidence on post quantity

Now we consider whether territory posting cost also influences the number of posts a territory receives. If users are sensitive to posting cost, then we expect that territory posting fee will be negatively correlated with number of posts. To test this, we run regressions of the following form:

$$\ln \text{Posts}_{jt} = \alpha \ln \text{Cost}_{jt} + \delta_j + \gamma_t + \epsilon_{jt} \tag{3}$$

where j indexes a territory and t indexes a week. Posts_{jt} is the number of posts posted in territory j in week t, Cost_{jt} is territory j's posting fee in week t (measured as the average across days), δ_j is a set of territory fixed effects, and γ_t is a set of week fixed effects. The results of this regression, for various combinations of fixed effects, are presented in Table 2. As expected, there is a robust and statistically significant negative relationship between territory posting cost and the quantity of posts in a territory. Using column 3 as our preferred specification, the results imply an elasticity of post quantity with respect to post cost of -0.265. This translates to: if the posting cost doubles, the territory can expect to receive 16.8 percent fewer posts per week.

4.2.1 Event study analysis

We now turn to an event study analysis to test whether there are systematic changes in post quantity prior to territory fee changes. We run the following regression:

$$\begin{aligned} \ln \text{Posts}_{jt} = &\sum_{s=-12}^{12} \alpha_s^{+} 1\left[\Delta \text{Cost}_{j,t-s} > 0\right] \\ &+ \sum_{s=-12}^{12} \alpha_s^{-} 1\left[\Delta \text{Cost}_{j,t-s} < 0\right] + \delta_j + \gamma_t + \epsilon_{ijt} \end{aligned} \tag{4}$$

The notation and logic is much the same as in (2), so we will not repeat ourselves here. Figure 4 shows the estimated values for $\alpha_s^{+/-}$ and their 95% confidence intervals.

Figure 4 shows that there is no systematic decrease in post quantity prior to a territory fee increase. If anything, there appears to be a systematic upward trend in post quantity prior to fee increases, followed by a downward trend in post quantity after the fee increase. For negative fee changes, there appears to be a slight systematic downward trend in post quantity prior to a fee decrease, followed by an upward trend in post quantity after a fee decrease.

Comparing Figures 3 and 4 reveals a pattern that is consistent with our hypothesized mechanism. Prior to a fee increase, post quantity trends upwards while post quality remains flat. The absence of a pre-trend in quality indicates that quality improvements observed after a fee increase are not the continuation of a pre-trend that was already underway. Rather, quality begins to rise only after the fee change takes effect, consistent with a cost-based screening mechanism. The rising quantity prior to a fee increase suggests that territory operators may be raising fees in response to growing volume, with the quality improvement emerging as a consequence of the higher fees screening out lower quality posts. The reverse pattern surrounding fee decreases tells a symmetric story. This story appears consistent with how some territory operators on the platform have described their rationale for fee changes. One territory operator is quoted saying:

> "For `science`, I only experimented very little at the beginning. Don't remember the specifics, but I must have tried really low (like 10-20 sats) and then a bit higher (like 100 sats). Because I had a personal policy of upvoting each post that had a modicum of quality (e.g. not just a link without context), I ended up with a small army of posters (assmilkers) that were doing the bare minimum but not enough to instigate comments or discussions. I think the reason for the 100 sats was to see how this increase would affect the willingness of those low-effort posters to post. It worked, but a side effect is that I was getting much less volume, also from quality posters. So I put it back down, and just let the community decide if a post is worth getting sats."

This operator's reasoning reflects awareness of the cost-based screening mechanism as a tool for managing post quantity and .

4.3 Discussion

Taken together, the results are both surprising and unsurprising. They are unsurprising in the sense that they conform with economic theory: demand curves slope downwards (when posting costs go up, number of posts goes down), and signaling theory works (when posting costs go up, higher quality posts are made). They are surprising in the sense that even such small micro-incentives (the average posting cost is just 51 sats, or about 5 cents) are enough to influence user behavior in such a way that post quality is improved. The results suggest that even very small posting fees can meaningfully affect the quantity-quality tradeoff in online discussion platforms.

5 Tip-based learning

We now consider our second hypothesis: that users learn from and respond to tip feedback by shifting their posting behavior towards better rewarded post types.

5.1 Model of post type choice

To test this hypothesis, we must first develop some notation for modeling user choice of post type. Consider a user i at time t, deciding to make a post from a finite number of post types, $c = 1, \ldots, C$. In theory, the post type could measure any number of different post attributes. In practice, and for parsimony's sake, we limit ourselves to a small number of types based primarily on the length of the post, whether it is a link-only post, and whether it includes hyperlinks and images.

Let V_{itc} be user i's utility for making a post of type c at time t. Let y_{itc} equal 1 if the user actually makes a type c post at time t, and 0 otherwise. The user chooses the post type that maximizes V_{itc}. Thus:

$$y_{itc} = 1\left[V_{itc} \geq V_{itc'}\ \forall c' = 1, \ldots, C\right] \tag{5}$$

We model the user's utility as follows:

$$V_{itc} = \alpha E\left[\ln z_{itc} | I_{itc}\right] + \delta_c + \gamma_c \cdot t + \epsilon_{itc} \tag{6}$$

where $E\left[\ln z_{itc} | I_{itc}\right]$ is the user's expected value of the ln of the sats they would receive on the post, conditional on their information set I_{itc}. δ_c is separate intercept for each post type, meant to reflect differences in baseline popularity of each post type. $\gamma_c \cdot t$ are post type specific linear time trends, which capture changes to the popularity of different post types over time which are not explainable by observed tip feedback. ϵ_{itc} is an error term which captures idiosyncratic features such as the user's intrinsic desire to make a type-c post in the moment. We model ϵ_{itc} as a Gumbel distribution, which turns the choice model into a standard multinomial logit choice model over C alternatives. Our hypothesis is that $\alpha > 0$; that is, the user's expected rewards for post type c will shift his decision-making towards a greater likelihood of making a type-c post.

5.2 Post types

To keep the model parsimonious, we categorize posts into five post types based on the number of words and the presence of images or hyperlinks in the post body. Table 3 shows the five types, their prevalence in the data, the average amount of sats earned in the first 48 hours, and the average number of replies received in the first 48 hours.

The first type of post is a link-only post with no words in the post body. Link-only posts are posts that link to an external URL with no additional text in the post body. Link-only posts earn the least amount of sats, and are generally seen by users as low effort posts because they contain no original content. However, link-only posts are the most common type of post on Stacker News.

The next four types of post are based on whether the post body contains above or below the median number of words, and whether the post body contains images or hyperlinks. Posts with more words tend to earn more sats on average than posts with less words, and posts with images or hyperlinks in the body earn more sats when combined with more words. Posts with more words may be associated with more effort and more original content, which may thereby be rewarded with more tips. Posts with images may be more visually appealing than posts without images, and thus also earn more sats. Hyperlinks in posts with many words are usually associated with linked references and citations in the post body, another sign of quality that may be accordingly rewarded.

Table 3 shows that posts with an above median number of words and with images or hyperlinks in the body are the best rewarded, by a significant margin. Posts with an above median number of words, but without images or hyperlinks in the body, are the second most rewarded type. The typical earnings of below median length posts are about the same with or without images and hyperlinks. And finally, link-only posts are the least rewarded type of post by a significant margin.

5.3 Information environment assumptions

It is not possible for us, as the researchers, to know the information set of each user in each moment that a choice was made. We therefore proceed by making different assumptions about the information environment and showing that the results are robust to the choice of assumption.

To start with, we calculate $E\left[\ln z_{itc}|I_{itc}\right]$ as follows:

$$E\left[\ln z_{itc}|I_{itc}\right] = \frac{\sum_{i'}\sum_{t'<t} y_{i't'c} w_{itc,i't'c} \ln z_{i't'c}}{\sum_{i'}\sum_{t'<t} y_{i't'c} w_{itc,i't'c}} \tag{7}$$

In other words, we calculate $E\left[\ln z_{itc}|I_{itc}\right]$ as the weighted average of the realized ln(zaps) for all type c posts made before time t (by any user). Variation is introduced across users and across times by means of the weights, which explicitly depend on both itc (the user making the decision) and $i't'c$ (the post from which information is being drawn).

As a baseline, we consider uniform weights as long as the post was made after the user became active. That is, $w_{itc,i't'c} = 1$ so long as t' is after user i's first post. This assumes that user i observes and draws information from all posts made from the time he became active to the current time.

As a second possibility, we consider weights that are based on recency. In this weighting scheme, $w_{itc,i't'c} = e^{-\lambda(t-t')}$, where λ is chosen for a half-life of 4 weeks. This information environment assumes that the user pays more attention to recent posts than to old posts, and variation is driven by the timing of the user's posts.

As a third possibility, we consider weights that are based on the proximity of the post to the user's peak activity hours. The idea of this weighting scheme derives from the fact that Stacker News is an international platform, and that different users are most active during different times of the day. For each user i, we measure s_i^*, the average time in the day (measured in seconds from midnight) that the user tends to post. Then, the weights are calculated as $w_{itc,i't'c} = \exp\left(-\frac{d(s_i^*,t')^2}{2\sigma^2}\right)$, where $d(s_i^*, t')$ measures the circular time-of-day distance between t' and s_i^*. This information environment assumes that users are more aware of posts made during their peak activity hours. σ^2 is chosen to generate a half-life of 3 hours.

5.4 Results

Table 4 shows the estimation results for this choice model under the aforementioned information assumptions. Columns 1-2 of Table 4 show the estimated coefficients under uniform weights, columns 3-4 show the coefficients under recency-based weights, and columns 5-6 shows the coefficients under proximity-to-peak-hours weights.

In the Table, "Observed Tips" is equivalent to $E\left[\ln z_{itc}|I_{itc}\right]$. In all six specifications shown, the coefficient on Observed Tips is estimated to be positive and statistically significant. This confirms our hypothesis that $\alpha > 0$; in other words, we find that users shift their posting behavior towards the post types that they observed to be better rewarded. This finding is robust to the various assumptions we made about the information environment.

In addition, we enrich the model by interacting Observed Tips with a measure of the user's experience level, measured by the natural log of the number of posts they've made. This interaction captures a natural feature of learning models which says that learning slows down as the user gets more experienced. When a user has many posts under their belt, they will already have a good sense of what works well for them and what doesn't, and observed tips on other posts are no longer as influential in their decision-making. Consistent with learning theory, we find a negative and statistically significant interaction between Observed Tips and the user's experience level. The finding is robust to our different choices of weighting functions.

Lastly, in columns 2, 4, and 6, we augment the model with a variable called "User

Excess Tips". "User Excess Tips" is equal to:

$$\text{User Excess Tips} = \frac{\sum_{t'<t} y_{it'c} w_{itc,it'c} \ln z_{it'c}}{\sum_{t'<t} y_{it'c} w_{itc,it'c}} - E\left[\ln z_{itc} | I_{itc}\right] \tag{8}$$

User Excess Tips measures the degree to which the user's own earnings on posts of type c, up to time t, have exceeded their estimate based on all observed posts. We expect the coefficient on User Excess Tips to be positive, because a user's own posts are expected to be more salient to their belief formation than other users' posts. A user's own posts could be more salient either because they pay more attention to their own posts, or because their excess earnings relative to other users conveys information about a post-type specific match quality. In either case, a user who earns more tips on a particular post type relative to other users is expected to shift their posting towards that post type. That is what we find, estimating a positive and statistically significant coefficient in all three specifications.

5.5 Discussion

In this section, we found evidence confirming our hypothesis that users respond to tip-driven feedback by shifting their posting behavior towards posts that appear to be better rewarded. Consistent with learning theory, we also found that new users are more responsive to prior observed tips than seasoned users, and that users are additionally responsive to excess tip feedback on their own posts. The findings were robust to various assumptions about the information environment.

If users shift their posting behavior towards better-rewarded types, then over time we should expect to see the share of those types increase in the aggregate, while the share of poorly rewarded types is expected to decrease. Figure 5 shows that this is indeed what we see. Link-only posts, the lowest effort and least rewarded type of post, decreased precipitously in the aggregate from the start of the sample period to the end. Non-link-only posts all increased, with above-median length posts with images or hyperlinks increasing in share the most. The findings suggest that tip-driven monetary feedback mechanisms could help to shape the overall quality of user submitted content, even if the monetary value of the tips is very small.

6 Profitability-based selection

We now test our last hypothesis: that consistently unprofitable users are more likely to exit the platform, whereas profitable users are more likely to stay. This would arise straightforwardly out of a learning model in which some users learn that their returns are consistently low.

To measure profitability, we only consider whether the user is able to earn enough tips and rewards to cover their posting costs. We do not consider their own tips to other users as part of their costs, since tipping is entirely optional. Moreover, some users may have altruistic motives to tip, as discussed in the introduction, so greater tipping could indicate a greater commitment to the platform.

We define a user as inactive if they go four or more consecutive weeks without making any posts, any comments, or zapping any items. The first week of inactivity is considered the week they went inactive. Inactive users can become active again, though we treat this event as exogenous.

The goal of this section is to estimate how user profitability affects the probability of becoming inactive. We estimate linear probability models of the following form:

$$\text{Inactive}_{it} = \alpha \text{Unprofitable}_{it} + \beta \text{Unprofitable}_{it} \times \text{Growth}_t + \gamma_t + \epsilon_{it} \tag{9}$$

where i indexes a user and t indexes a week. Inactive_{it} is a binary indicator for whether user i became inactive on week t, Unprofitable_{it} is a binary indicator for whether the user was unprofitable in the last 8 weeks, and Growth_t is the percentage growth in Bitcoin-USD price in the last 8 weeks. γ_t is a week fixed effect to capture global time trends, and ϵ_{it} is the error term. We hypothesize that $\alpha > 0$, indicating that users who have been unprofitable are more likely to become inactive, and that $\beta > 0$, that this effect is amplified when Bitcoin's price has recently appreciated.

Table 5 reports the results for this regression. Column 1 shows the result when only profitability in the last 8 weeks is included as a regressor. Column 2 additionally controls for the number of items the user posted in the last 8 weeks. Unsurprisingly, users who have posted more items in the last 8 weeks are less likely to become inactive, as it shows a greater commitment to the platform. Including the number of items in the last 8 weeks significantly moderates the effect of unprofitability, suggesting that there are some unprofitable users who post a lot anyway. Column 3 adds week fixed effects to control for global time trends, but the coefficient of interest is not much changed. Lastly, column 4 interacts unprofitability with the Bitcoin price appreciation over the last 8 weeks.

Using column 4 as the preferred specification, the results show that users who were unprofitable in the last 8 weeks are 5.5 percentage points more likely to become inactive than users who were profitable in the last 8 weeks. This is in comparison to a baseline inactive probability of 14.1% per week. In addition, column 4 shows that unprofitable users are even more likely to become inactive when the Bitcoin price has recently appreciated. If the Bitcoin price has appreciated by 10 percent in the last 8 weeks, this increases the effect of unprofitability from a 5.5 percentage-point to a 6.1 percentage-point increased probability of exit. This suggests that users are sensitive to not just their profitability as measured in sats, but also their profitability as measured in purchasing power.

7 Conclusion

Our analysis of Stacker News data confirms three central hypotheses: first, that posting costs can increase the quality of posted content by screening out lower-quality content, consistent with signaling theory; second, that the tip-based feedback shapes the quality and type of posts that users make; and third, that unprofitable users are more likely to leave the platform while profitable users are more likely to stay. This third point suggests that content quality may further increase over time as users who consistently produce low-value content eventually exit the platform.

More broadly, the results suggest that users of digital platforms are responsive to financial micro-incentives, even ones of very small monetary value. Integrating micropayment systems into user-generated content platforms could therefore be an effective approach to managing the quantity-quality tradeoff in content production. Cost-based screening may become especially relevant as platforms seek to discourage AI-generated content submissions. Our results may therefore offer practical guidance to platform designers and regulators considering the integration of direct payment features.

References

Andreoni, J. (1990). Impure altruism and donations to public goods: A theory of warm-glow giving. *The Economic Journal 100*(401), 464–477.

Aridor, G., R. Jiménez-Durán, R. Levy, and L. Song (2024). The economics of social media. *Journal of Economic Literature 62*(4), 1422–1474.

Azar, O. H. (2020). The economics of tipping. *Journal of Economic Perspectives 34*(2), 215–236.

Conley, T. G. and C. R. Udry (2010). Learning about a new technology: Pineapple in ghana. *American Economic Review 100*(1), 35–69.

Crumpler, H. and P. J. Grossman (2008). An experimental test of warm glow giving. *Journal of Public Economics 92*(5), 1011–1021.

El-Komboz, L. A., A. Kerkhof, and J. Loh (2023). Platform partnership programs and content supply: Evidence from the Youtube "Adpocalypse". CESifo working paper no. 10363.

Joseph, K. and A. Thevaranjan (2008). Investigating pricing solutions to combat spam: Postage stamp and bonded senders. *Journal of Interactive Marketing 22*(1), 21–35.

Kerkhof, A. (2024). Advertising and content differentiation: Evidence from youtube. *The Economic Journal 134*(663), 2912–2950.

Kraut, R. E., P. Resnick, S. Kiesler, M. Burke, Y. Chen, N. Kittur, J. Konstan, Y. Ren, and J. Riedl (2012). *Building Successful Online Communities: Evidence-Based Social Design*. The MIT Press.

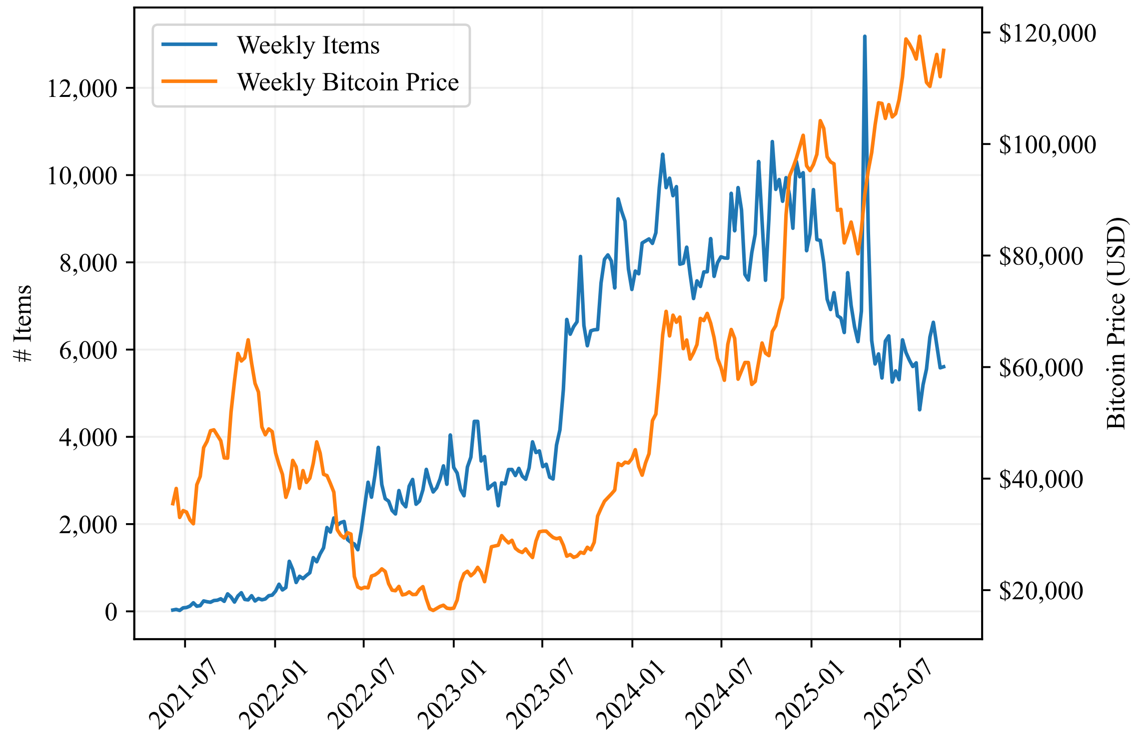
Luca, M. (2015). User-generated content and social media. In S. P. Anderson, J. Waldfogel, and D. Strömberg (Eds.), *Handbook of Media Economics*, pp. 563–592.

Mak, V., R. Zwick, A. R. Rao, and J. A. Pattaratanakun (2015). "Pay what you want" as threshold public good provision. *Organizational Behavior and Human Decision Processes 127*, 30–43.

Sun, M. and F. Zhu (2013). Ad revenue and content commercialization: Evidence from blogs. *Management Science 59*(10), 2314–2331.

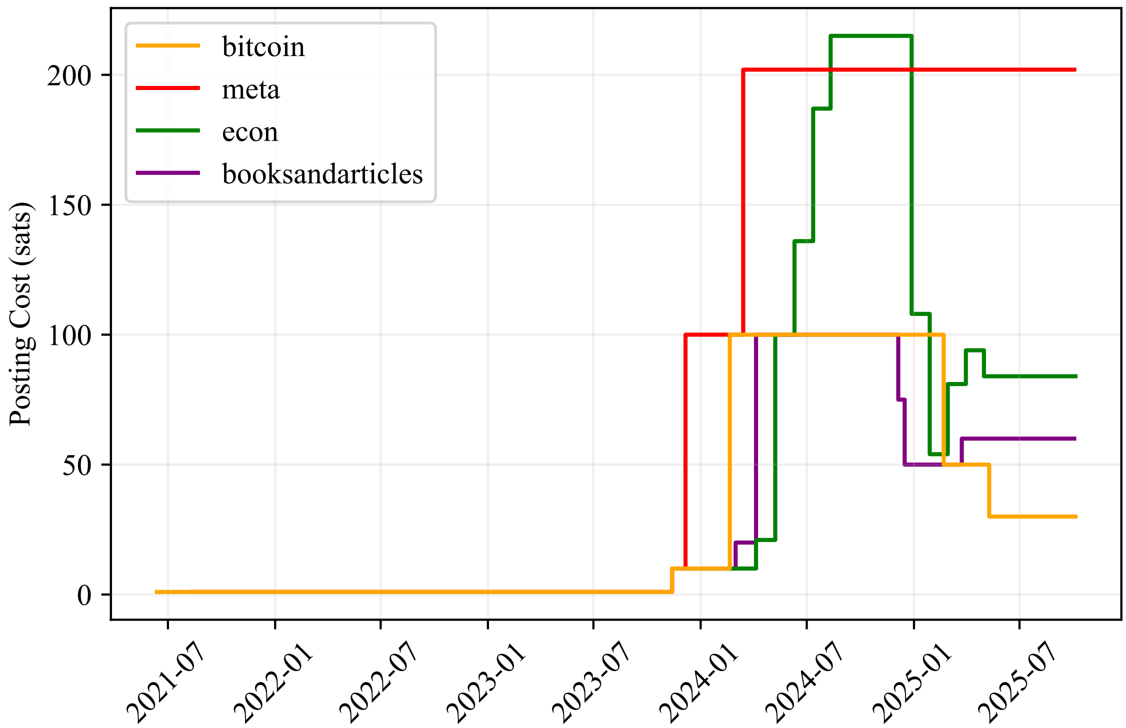
Tchernichovski, O., L. C. Parra, D. Fimiarz, A. Lotem, and D. Conley (2019). Crowd wisdom enhanced by costly signaling in a virtual rating system. *Proceedings of the National Academy of Sciences 116*(15), 7256–7265.

Figure 1: Weekly Stacker News Items and Bitcoin Price



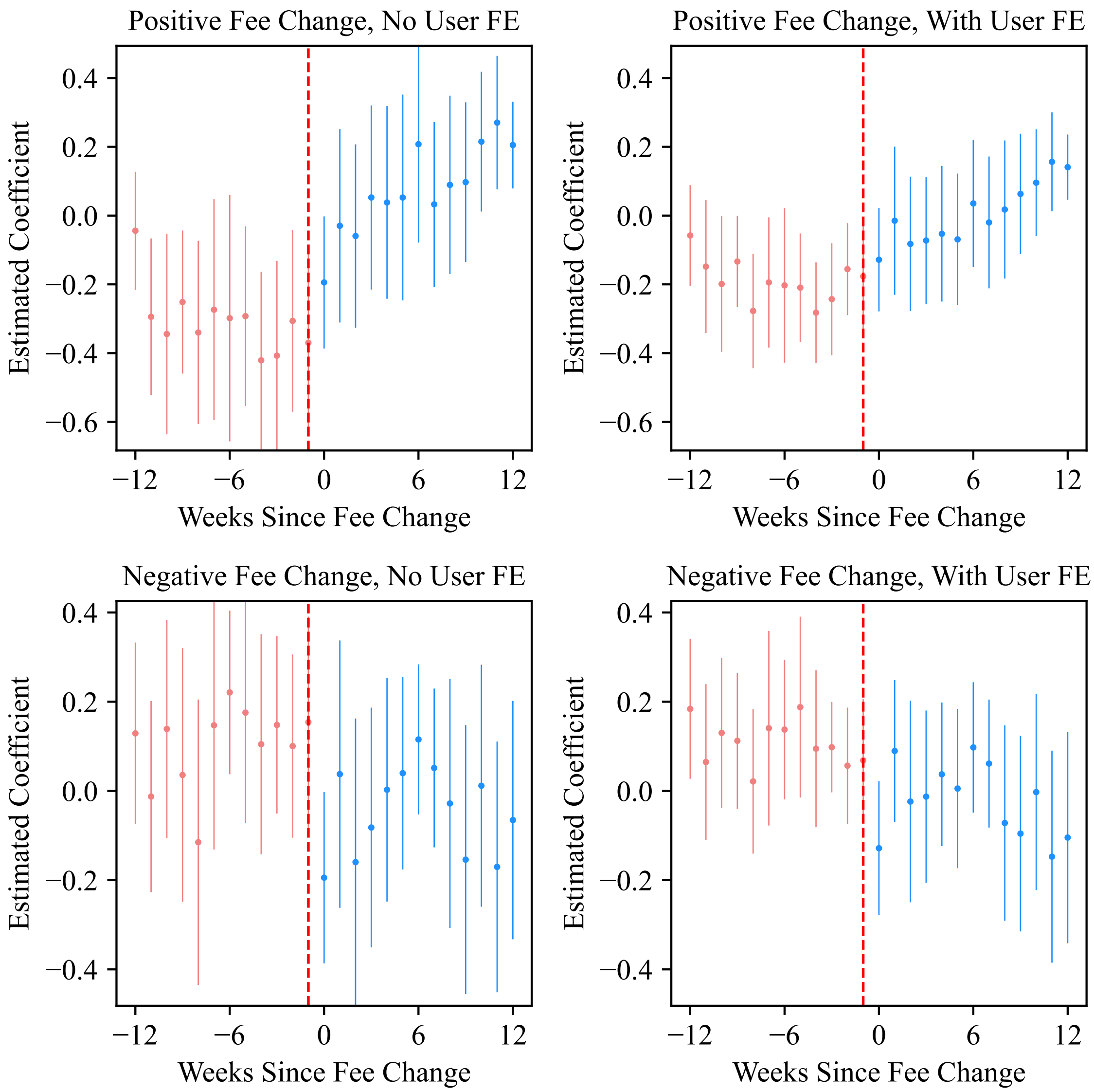
Note: Shows the number of items posted on Stacker News by week along with the weekly Bitcoin price. Weekly Bitcoin price is calculated as the average of the midpoint between daily highs and lows as reported by CoinMarketCap.com.

Figure 2: Posting Cost Histories for Select Territories



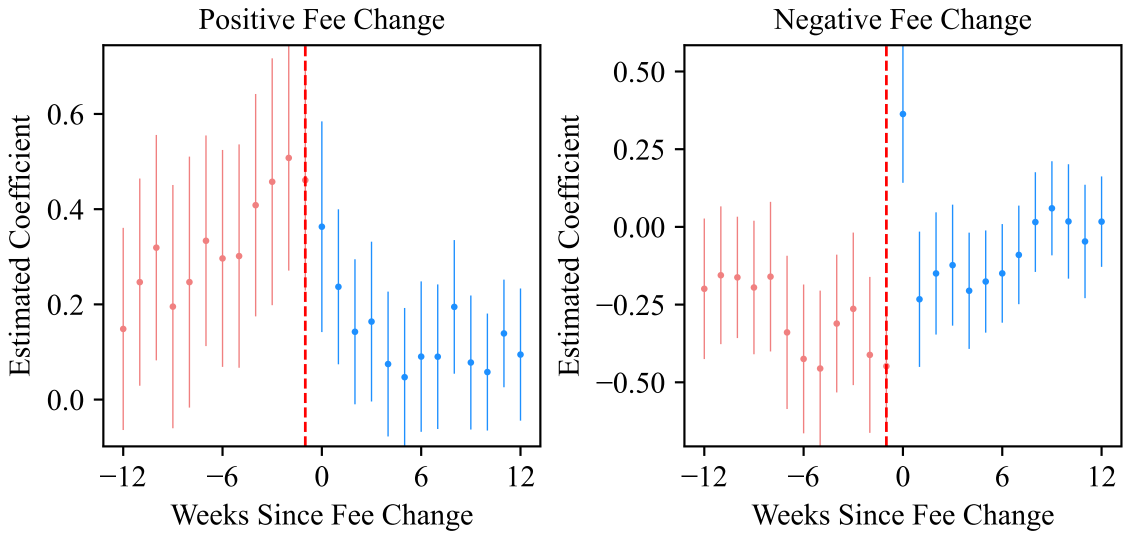
Note: This figure shows the territory fee path for four selected territories. The territories were selected both for their popularity and to highlight the differences in timing and magnitude of fee changes across territories.

Figure 3: Post Quality Event Study



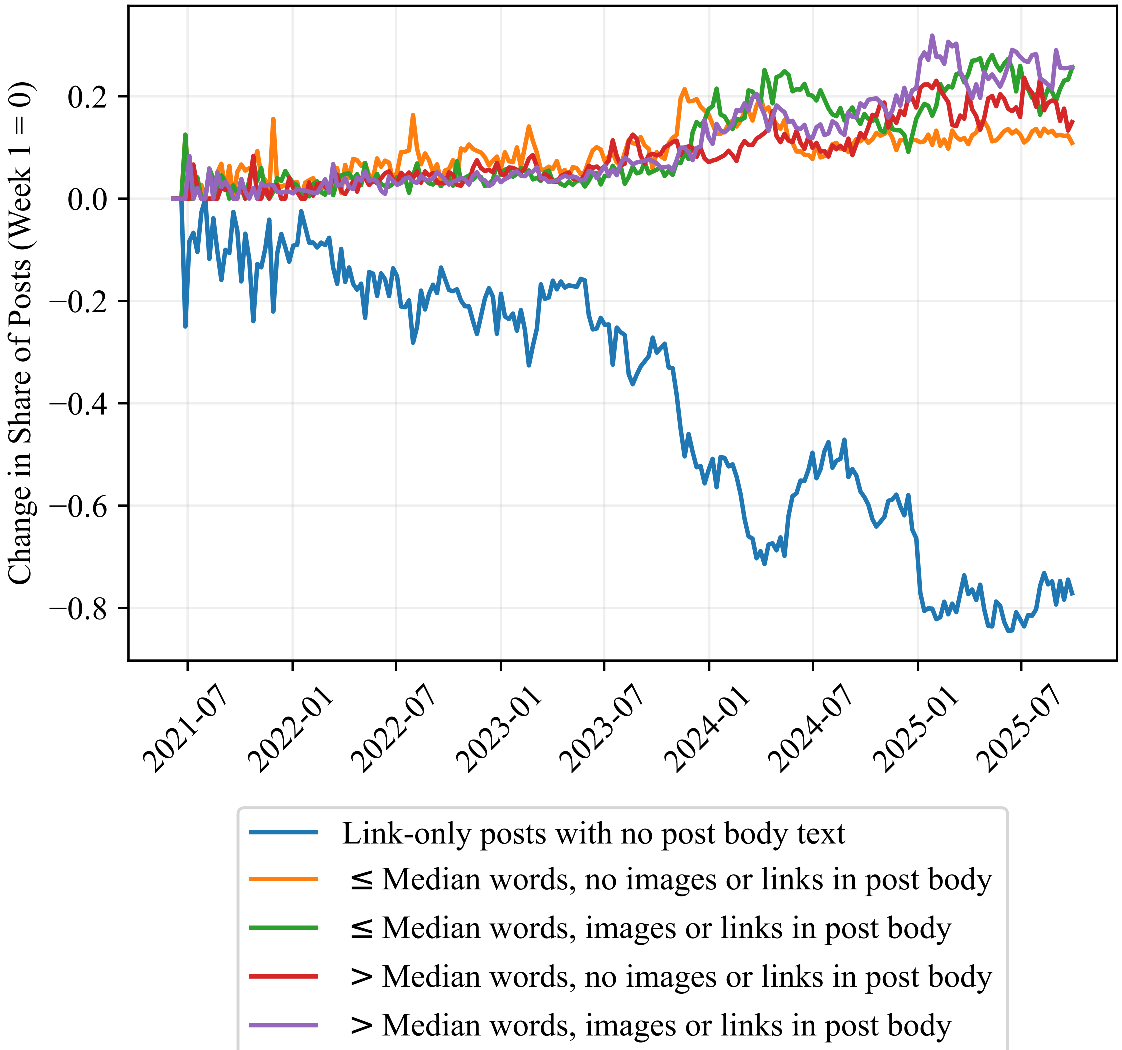
Note: Shows estimated coefficients and 95% confidence intervals from the post quality event study regression specified by equation (2). All regressions shown include territory-owner fixed effects and week fixed effects.

Figure 4: Post Quantity Event Study



Note: Shows estimated coefficients and 95% confidence intervals from the post quantity event study regression specified by equation (4). All regressions shown include territory fixed effects and week fixed effects.

Figure 5: Change in Post Composition Over Time



Note: Shows how the share of each type of post changed since the first week. See Section **??** for details.

Table 1: Effect of Posting Cost on Post Quality

	Dependent variable: ln(Zaps)			
	(1)	(2)	(3)	(4)
ln(Posting Cost)	0.393***	0.273***	0.219***	0.091**
	(0.048)	(0.058)	(0.050)	(0.044)
Constant	2.530***			
	(0.076)			
Territory FE	N	Y	Y	Y
Week FE	N	Y	Y	Y
Territory Operator FE	N	N	Y	Y
User FE	N	N	N	Y
Observations	179,480	179,477	179,476	178,197
R^2	0.079	0.181	0.184	0.402

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Regression of number of sats a post earns in the first 48 hours on posting cost. The unit of observation is a post. Standard errors are clustered by territory.

Table 2: Effect of Posting Cost on Post Quantity

	Dependent variable: ln(Posts)		
	(1)	(2)	(3)
ln(Posting Cost)	-0.242***	-0.266***	-0.265***
	(0.079)	(0.063)	(0.074)
Constant	3.328***		
	(0.415)		
Territory FE	N	Y	Y
Week FE	N	Y	Y
Territory Operator FE	N	N	Y
Observations	6,023	6,020	6,020
R^2	0.092	0.776	0.788

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Regression of number of posts in a territory on posting costs for that week. The unit of observation is a territory-week. Standard errors are clustered by territory.

Table 3: Post Types Summary

	Number of posts	Avg. sats received in first 48 hours	Avg. replies received in first 48 hours
Link-only posts with no post body text	90,154	167	1.3
≤Median words, no images or links in post body	19,385	296	5.1
≤Median words, images or links in post body	23,517	271	3.1
>Median words, no images or links in post body	19,172	564	6.8
>Median words, images or links in post body	23,629	1,112	6.9

Notes: This table reports the number of posts and the average amount of sats and replies received in the first 48 hours, for each post type.

Table 4: Effect of Observed Tip Feedback on Post Type Choice

	Observed feedback weighted by:					
	Uniform		Recency		Distance to user's peak activity	
	(1)	(2)	(3)	(4)	(5)	(6)
Observed Tips	0.844***	0.905***	0.980***	0.991***	0.819***	0.881***
	(0.018)	(0.018)	(0.017)	(0.017)	(0.016)	(0.017)
Observed Tips × User's Experience	-0.187***	-0.181***	-0.207***	-0.206***	-0.177***	-0.173***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
User Excess Tips		0.123***		0.019***		0.117***
		(0.003)		(0.003)		(0.003)
Post Type FE	Y	Y	Y	Y	Y	Y
Post Type × Linear Time Trend	Y	Y	Y	Y	Y	Y
Observations	166,187	166,187	166,187	166,187	166,187	166,187
McFadden's Pseudo-R^2	0.209	0.212	0.214	0.215	0.209	0.212

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This table reports estimated coefficients for the conditional logit model of post type choice described in Section 5 equation (**??**). Each unit of observation is a user i's choice to make a post of type c at time t. The user chooses the post type based on their prior observation of tips earned by posts of each type. "Observed Tips" is a weighted average of ln(sats) earned on posts of each type c prior to time t. See Section 5 for a description of the model and weighting functions. "User's Experience" is the ln of the number of posts the user i has made up to time t. "User Excess Tips" is the weighted average of the user's own ln(sats) for each post type c earned in excess of the overall weighted average for that post type.

Table 5: Effect of Profitability on User Exit

	Dependent variable: Whether user becomes inactive			
	(1)	(2)	(3)	(4)
Unprofitable in last 8 weeks	0.105***	0.062***	0.060***	0.055***
	(0.003)	(0.003)	(0.003)	(0.003)
... × BTC price appreciation in last 8 weeks				0.056***
				(0.014)
log(Items posted in last 8 weeks)		-0.029***	-0.029***	-0.029***
		(0.001)	(0.001)	(0.001)
Constant	0.079***	0.157***		
	(0.001)	(0.002)		
Week FE	N	N	Y	Y
Observations	92,956	92,956	92,956	92,956
R^2	0.018	0.042	0.048	0.048

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Linear probability model of whether a user becomes inactive in a given week. A user becomes inactive if they start a four or more week consecutive period of no activity. The unit of observation is a user-week.